MRC Lecture 3 — Derivation Guide 3

Numerical InfoNCE Example: A Complete Worked Computation

Sai T. Reddy — ETH Zurich / BIIE

1. Motivation

Derivation Guides 1 and 2 derived the InfoNCE loss and the symmetric loss from the convergence equation. This guide works through a complete numerical example with a batch of $N=4$ pairs, computing every number explicitly. The goal is to build intuition for what the loss function "sees" and how it drives the embeddings to organize correctly.

2. Setup

Batch. Four antibody-antigen binding pairs:

- Pair 1: Antibody 1 binds Antigen 1
- Pair 2: Antibody 2 binds Antigen 2
- Pair 3: Antibody 3 binds Antigen 3
- Pair 4: Antibody 4 binds Antigen 4

Similarity matrix. After encoding and computing cosine similarities divided by temperature τ, suppose the 4×4 similarity matrix is:

$$S=\begin{pmatrix} 2.0 & 0.5 & 0.3 & 0.1 \\ 0.4 & 1.8 & 0.6 & 0.2 \\ 0.2 & 0.3 & 2.2 & 0.5 \\ 0.1 & 0.4 & 0.3 & 1.9 \end{pmatrix}$$

The diagonal entries ($S_{11}=2.0$, $S_{22}=1.8$, $S_{33}=2.2$, $S_{44}=1.9$) are the scores for correct binding pairs. The off-diagonal entries are scores for non-binding (negative) pairs. The diagonal entries are larger than the off-diagonal entries — the model has partially learned to discriminate correct from incorrect pairs, but not perfectly.

3. Agent→Target Loss (Row-Wise InfoNCE)

For each agent (row), compute the probability of selecting the correct target.

Row 1 (Antibody 1):

Step 1. Exponentiate all entries in row 1:

$$\exp(2.0)=7.389, \exp(0.5)=1.649, \exp(0.3)=1.350, \exp(0.1)=1.105$$

Step 2. Compute the partition function (sum):

$$Z_1=7.389+1.649+1.350+1.105=11.493$$

Step 3. Probability of the correct target (Antigen 1):

$$P(1 / \mathrm{Ab}_1)=\frac{7.389}{11.493}=0.643$$

Step 4. Negative log-probability:

$$-\log P(1 / \mathrm{Ab}_1)=-\log(0.643)=0.442$$

Row 2 (Antibody 2):

$$\exp(0.4)=1.492, \exp(1.8)=6.050, \exp(0.6)=1.822, \exp(0.2)=1.221$$

$$Z_2 = 1.492 + 6.050 + 1.822 + 1.221 = 10.585$$
$$P(2 / \mathrm{Ab}_2) = \frac{6.050}{10.585} = 0.572$$
$$-\log P(2 / \mathrm{Ab}_2) = -\log(0.572) = 0.558$$

Row 3 (Antibody 3):

$$\exp(0.2) = 1.221, \exp(0.3) = 1.350, \exp(2.2) = 9.025, \exp(0.5) = 1.649$$
$$Z_3 = 1.221 + 1.350 + 9.025 + 1.649 = 13.245$$
$$P(3 / \mathrm{Ab}_3) = \frac{9.025}{13.245} = 0.681$$
$$-\log P(3 / \mathrm{Ab}_3) = -\log(0.681) = 0.384$$

Row 4 (Antibody 4):

$$\exp(0.1) = 1.105, \exp(0.4) = 1.492, \exp(0.3) = 1.350, \exp(1.9) = 6.686$$
$$Z_4 = 1.105 + 1.492 + 1.350 + 6.686 = 10.633$$
$$P(4 / \mathrm{Ab}_4) = \frac{6.686}{10.633} = 0.629$$
$$-\log P(4 / \mathrm{Ab}_4) = -\log(0.629) = 0.464$$

Agent→target loss:

$$L_{\text{agent} \to \text{target}} = \frac{1}{4}(0.442 + 0.558 + 0.384 + 0.464) = \frac{1.848}{4} = 0.462$$

---

4. Target→Agent Loss (Column-Wise InfoNCE)

For each target (column), compute the probability that the correct agent selects it.

Column 1 (Antigen 1):

$$\exp(S_{11}) = 7.389, \exp(S_{21}) = 1.492, \exp(S_{31}) = 1.221, \exp(S_{41}) = 1.105$$
$$Z_1^{\text{col}} = 7.389 + 1.492 + 1.221 + 1.105 = 11.207$$
$$P(\mathrm{Ab}_1 / \mathrm{Ag}_1) = \frac{7.389}{11.207} = 0.659$$
$$-\log(0.659) = 0.417$$

Column 2 (Antigen 2):

$$\exp(S_{12}) = 1.649, \exp(S_{22}) = 6.050, \exp(S_{32}) = 1.350, \exp(S_{42}) = 1.492$$
$$Z_2^{\text{col}} = 1.649 + 6.050 + 1.350 + 1.492 = 10.541$$
$$P(\mathrm{Ab}_2 / \mathrm{Ag}_2) = \frac{6.050}{10.541} = 0.574$$
$$-\log(0.574) = 0.555$$

Column 3 (Antigen 3):

$$\exp(S_{13}) = 1.350, \exp(S_{23}) = 1.822, \exp(S_{33}) = 9.025, \exp(S_{43}) = 1.350$$
$$Z_3^{\text{col}} = 1.350 + 1.822 + 9.025 + 1.350 = 13.547$$
$$P(\mathrm{Ab}_3 / \mathrm{Ag}_3) = \frac{9.025}{13.547} = 0.666$$
$$-\log(0.666) = 0.406$$

Column 4 (Antigen 4):

$$\exp(S_{14}) = 1.105, \exp(S_{24}) = 1.221, \exp(S_{34}) = 1.649, \exp(S_{44}) = 6.686$$

$$Z_4^{\text{col}} = 1.105 + 1.221 + 1.649 + 6.686 = 10.661$$

$$P(\text{Ab}_4 / \text{Ag}_4) = \frac{6.686}{10.661} = 0.627$$

$$-\log(0.627) = 0.467$$

Target→agent loss:

$$L_{\text{target} \to \text{agent}} = \frac{1}{4}(0.417 + 0.555 + 0.406 + 0.467) = \frac{1.845}{4} = 0.461$$

---

5. Symmetric Loss

$$L_{\text{symmetric}} = \frac{1}{2}(0.462 + 0.461) = 0.462$$

In this example, the two directions give nearly identical losses (0.462 vs. 0.461), indicating the embedding space is approximately balanced between modalities.

---

6. Comparison to Theoretical Bounds

Scenario	Loss value	Meaning
Random guessing	$\log 4 = 1.386$	All targets equally likely; no discrimination
This example	0.462	Partial discrimination; correct targets preferred but not dominant
Perfect retrieval	0	Correct target has probability 1; perfect discrimination

The model's loss (0.462) is well below the random baseline (1.386), indicating substantial learning, but well above the theoretical minimum (0), indicating room for improvement. Training proceeds by gradient descent to push the loss toward 0.

---

7. What Gradient Descent Does

The gradient signal. The InfoNCE gradient pushes the model to:

1. **Increase** the diagonal entries S_{ii} — make correct pairs more similar
2. **Decrease** the off-diagonal entries S_{ij} $(j \neq i)$ — make incorrect pairs less similar

Concretely, for Antibody 2 (which had the highest loss at 0.558, corresponding to the lowest correct-pair probability at 0.572):

- The gradient increases $S_{22} = 1.8$ (push Antibody 2 and Antigen 2 closer)
- The gradient decreases $S_{23} = 0.6$ (the largest off-diagonal entry in row 2 — the hardest negative for Antibody 2)

The model focuses its learning on the hardest cases — the pairs where the correct target is least dominant. This is the contrastive learning analogue of the immune system's affinity maturation: selection pressure is strongest where discrimination is weakest.

---

8. Early Training vs. Late Training

Early training (untrained model):

Suppose all entries of S are approximately equal (the model has not yet learned to discriminate):

$$S_{\text{early}} \approx \begin{pmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \end{pmatrix}$$

Every target has equal probability: $P(i / \text{Ab}_i) = 1/4 = 0.25$ for all i.
Loss: $-\log(0.25) = \log 4 = 1.386$.

Late training (well-trained model):

Suppose the diagonal dominates:

$$S_{\text{late}} \approx \begin{pmatrix} 5.0 & 0.1 & 0.1 & 0.1 \\ 0.1 & 5.0 & 0.1 & 0.1 \\ 0.1 & 0.1 & 5.0 & 0.1 \\ 0.1 & 0.1 & 0.1 & 5.0 \end{pmatrix}$$

For row 1: $\exp(5.0) = 148.4$, $\exp(0.1) = 1.105$. $Z = 148.4 + 3(1.105) = 151.7$.
$P(1 / \text{Ab}_1) = 148.4 / 151.7 = 0.978$.
Loss per row: $-\log(0.978) = 0.022$.
Total loss: ≈ 0.022.
The loss has decreased from 1.386 (random) to 0.022 (near-perfect retrieval) as the model learned to assign high similarity to correct pairs and low similarity to incorrect pairs.

9. Summary

Quantity	Value in this example
Batch size N	4
Agent→target loss	0.462
Target→agent loss	0.461
Symmetric loss	0.462
Random baseline	$\log 4 = 1.386$
Best possible	0
Best pair (highest P)	Ab 3 → Ag 3 ($P = 0.681$)
Worst pair (lowest P)	Ab 2 → Ag 2 ($P = 0.572$)

10. Checkpoint Questions

Q1. In the example, Antibody 3 has the highest correct-pair probability (0.681) and the highest diagonal score ($S_{33} = 2.2$). But Antibody 3 also has a relatively high off-diagonal entry ($S_{34} = 0.5$). Why doesn't this hurt more?
Answer: The probability depends on the diagonal score relative to ALL off-diagonal scores, not just the largest one. Antibody 3's diagonal ($S_{33} = 2.2$) is much larger than any of its off-diagonal entries (0.2, 0.3, 0.5). The partition function $Z_3 = 13.245$ is dominated by $\exp(2.2) = 9.025$. Even though $S_{34} = 0.5$ is the largest off-diagonal entry, its exponential (1.649) is small compared to the diagonal's exponential (9.025). The exponential amplification from softmax means that small score advantages become large probability advantages.
Q2. If you doubled the batch size from $N = 4$ to $N = 8$ (by adding 4 more pairs), what would happen to the loss value, assuming the new pairs are random negatives?
Answer: The loss would increase because there are more competitors in the denominator. With $N = 4$, each correct pair competes against 3 negatives. With $N = 8$, each competes against 7 negatives. The denominator grows, the probability of the correct target decreases, and the negative log-probability

(the loss) increases. The theoretical maximum also increases from $\log 4 = 1.386$ to $\log 8 = 2.079$. Larger batches provide harder discrimination tasks and richer learning signals.

Q3. In §7, we noted that the gradient focuses on the "hardest" cases. Which antibody-antigen pair would receive the strongest gradient update, and why?

Answer: Antibody 2–Antigen 2, because it has the lowest correct-pair probability (0.572) and therefore the highest per-pair loss (0.558). The gradient of InfoNCE with respect to the similarity score S_{22} is proportional to $\left(1 - P\left(2 / \mathrm{Ab}_2\right)\right) = 1 - 0.572 = 0.428$. For Antibody 3, the gradient is $\left(1 - 0.681\right) = 0.319$ — weaker because the model is already more confident about this pair. InfoNCE automatically allocates more learning signal to the pairs where the model is least certain.

11. Connection to Future Lectures

- **Lecture 7:** The SFM architecture prescribes the symmetric InfoNCE loss as the training objective. The numerical example here illustrates what the model optimizes during training.
- **Lecture 8:** The Level 1/2/3 classification predicts that the loss decreases faster (fewer training steps to reach low loss) when the architecture implements the governing equation (Level 3). The progression from random ($L = \log N$) to trained ($L \approx 0$) shown in §8 is faster for Level 3 architectures because the model performs parameter estimation rather than function approximation.
- **Lecture 9:** The MRC scaling law predicts how the converged loss (or equivalently, retrieval accuracy) improves as a function of training data diversity. The numerical example here shows one batch; the scaling law predicts how performance changes as the number of unique binding relationships in the training set increases.